

PROBLEMS SUBMIT CODE MY SUBMISSIONS STATUS HACKS ROOM STANDINGS CUSTOM INVOCATION

C. DZY Loves Fibonacci Numbers

time limit per test: 4 seconds
memory limit per test: 256 megabytes

In mathematical terms, the sequence F_n of Fibonacci numbers is defined by the recurrence relation

$$F_1 = 1; F_2 = 1; F_n = F_{n-1} + F_{n-2} \ (n > 2).$$

DZY loves Fibonacci numbers very much. Today DZY gives you an array consisting of n integers: $a_1, a_2, \dots, a_n$. Moreover, there are m queries, each query has one of the two types:

- Format of the query "1 $l\ r$ ". In reply to the query, you need to add F_{i-l+1} to each element a_i , where $l \leq i \leq r$.
- Format of the query "2 $l\ r$ ". In reply to the query you should output the value of $\sum_{i=l}^r a_i$ modulo 1000000009 ($10^9 + 9$).

Help DZY reply to all the queries.

Input

The first line of the input contains two integers n and m ($1 \leq n, m \leq 300000$). The second line contains n integers $a_1, a_2, \dots, a_n$ ($1 \leq a_i \leq 10^9$) — initial array a .

Then, m lines follow. A single line describes a single query in the format given in the statement. It is guaranteed that for each query inequality $1 \leq l \leq r \leq n$ holds.

Output

For each query of the second type, print the value of the sum on a single line.

Examples

input	Copy
4 4 1 2 3 4 1 1 4 2 1 4 1 2 4 2 1 3	
output	Copy
17 12	

Note

After the first query, $a = [2, 3, 5, 7]$.

For the second query, $sum = 2 + 3 + 5 + 7 = 17$.

After the third query, $a = [2, 4, 6, 9]$.

For the fourth query, $sum = 2 + 4 + 6 = 12$.

Codeforces Round #FF (Div. 1)

Finished

Practice

Virtual participation

Virtual contest is a way to take part in past contest, as close as possible to participation on time. It is supported only ICPC mode for virtual contests. If you've seen these problems, a virtual contest is not for you - solve these problems in the archive. If you just want to solve some problem from a contest, a virtual contest is not for you - solve this problem in the archive. Never use someone else's code, read the tutorials or communicate with other person during a virtual contest.

Start virtual contest

Clone Contest to Mashup

You can clone this contest to a mashup.

Clone Contest

Submit?

Language: GNU G++23 14.2 (64 bit, ms)

Choose file: Choose File No file chosen

Submit

Submission	Time	Verdict
332802149	Aug/07/2025 17:30	Accepted
332715878	Aug/07/2025 04:01	Accepted
332714958	Aug/07/2025 03:37	Time limit exceeded on test 10
332714828	Aug/07/2025 03:33	Wrong answer on test 4
293110404	Nov/24/2024 17:37	Memory limit exceeded on test 8
293105857	Nov/24/2024 17:02	Memory limit exceeded on test 8
293104946	Nov/24/2024 16:55	Wrong answer on test 5
293104034	Nov/24/2024 16:49	Wrong answer on test 5
293103138	Nov/24/2024 16:42	Memory limit exceeded on test 8
293102807	Nov/24/2024 16:40	Time limit exceeded on test 8

Problem tags

data structures

math

number theory

*2400

No tag edit access

Contest materials